UNIT 2 – Data Storage Technologies (Hadoops,HDFS, MapReduce)

Task 1: Compare Traditional DB vs. NoSQL/HDFS
Answer : Traditional databases store data in a structured format, usually in tables containing rows and columns.
NoSQL databases store structured, semi-structured, and unstructured data.
HDFS is a distributed file system that stores large files across multiple machines in a Hadoop cluster.

Comparison table

Feature	Traditional Database	NoSQL	HDFS
Scalability	Mostly vertical	Horizontal	Horizontal
Replication	Database replication	Built-in replication in many systems	Block replication
Performance	Fast for transactions	Fast for large-scale data access	High-throughput processing
The types of data handled by each system	Student records, Employee details, Banking, transactions, Customer, information	JSON documents, Social media data, Product catalogs, Sensor data	Large text files, Images and videos, Log files, Big data datasets, Machine-generated data
Suitable use cases	Banking, school management, billing systems	Social media applications, e-commerce, real-time applications	Big data analytics, log processing, large-scale data storage

Task 2:
A. Install & Configure Hadoop
A installation hadoop:

Creating a new group 'bdagirls' and adding a user the user is 'students'.

sodu addgroup bdagirls

sudo adduser --ingroup bdagirls students

```
mysql> EXIT
Bye
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ cat /proc/sys/net/ipv6/conf/all/disable_ipv6
0
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo addgroup bdagirls
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo adduser --ingroup bdagirls students
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
Changing the user information for students
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo mkdir -p /usr
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

creating a diretroy /usr **sudo mkdir -p /usr**
moving hadoop jar file into /usr and untar hadoop, command:**tar -xvzf hadoop-3.3.6.tar.gz -C /usr**
raname file name to 'hadoop' command is **sudo mv /usr/hadoop-3.3.6 /usr/hadoop**

```
drwxr-xr-x 244 root   root       12288 Oct 30 18:15 share
drwxr-xr-x   8 root   root        4096 Oct 30 18:09 src
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo mv /usr/hadoop-3.3.6 /usr/hadoop
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ ls -l /usr
total 713144
drwxr-xr-x   2 root   root       57344 Oct 30 18:15 bin
drwxr-xr-x   2 root   root        4096 Aug 26 10:59 games
drwxr-xr-x  10 nitttr nitttr      4096 Jun 18  2023 hadoop
-rw-r--r--   1 nitttr nitttr 730107476 Sep 25 12:24 hadoop-3.3.6.tar.gz
drwxr-xr-x  42 root   root       16384 Oct 30 18:14 include
drwxr-xr-x  98 root   root        4096 Oct 30 18:15 lib
drwxr-xr-x   2 root   root        4096 Oct 28 06:08 lib64
drwxr-xr-x  29 root   root       12288 Oct 30 18:09 libexec
drwxr-xr-x  11 root   root        4096 Aug 26 10:57 local
drwxr-xr-x   2 root   root       20480 Oct 30 18:15 sbin
drwxr-xr-x 244 root   root       12288 Oct 30 18:15 share
drwxr-xr-x   8 root   root        4096 Oct 30 18:09 src
```

Changing ownership of hadoop to 'students' in 'bdagirls'.
sudo chown -R student:bdagirls /usr/hadoop

```
chown: invalid group: 'students:bdagirls'
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo chown -R students:bdagirls /usr/hadoop
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ ls -l
total 36
drwxr-xr-x 2 nitttr nitttr 4096 Oct 27 14:17 Desktop
drwxr-xr-x 2 nitttr nitttr 4096 Oct 30 18:36 Documents
drwxr-xr-x 2 nitttr nitttr 4096 Oct 30 18:31 Downloads
drwxr-xr-x 2 nitttr nitttr 4096 Oct 27 14:17 Music
drwxr-xr-x 3 nitttr nitttr 4096 Oct 30 18:25 Pictures
drwxr-xr-x 2 nitttr nitttr 4096 Oct 27 14:17 Public
drwxr-xr-x 2 nitttr nitttr 4096 Oct 27 14:17 Templates
drwxr-xr-x 2 nitttr nitttr 4096 Oct 27 14:17 Videos
drwx------ 6 nitttr nitttr 4096 Oct 30 18:25 snap
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ ls -l /usr
total 713144
drwxr-xr-x   2 root     root         57344 Oct 30 18:15 bin
drwxr-xr-x   2 root     root          4096 Aug 26 10:59 games
drwxr-xr-x  10 students bdagirls      4096 Jun 18  2023 hadoop
-rw-r--r--   1 nitttr   nitttr   730107476 Sep 25 12:24 hadoop-3.3.6.tar.gz
drwxr-xr-x  42 root     root         16384 Oct 30 18:14 include
drwxr-xr-x  98 root     root          4096 Oct 30 18:15 lib
drwxr-xr-x   2 root     root          4096 Oct 28 06:08 lib64
```

moving to 'students' user **su students** and
creating a ssh key generate for password less connection and copying the fingerprint to authorized keys, and test a secure passwordless connetion
ssh-keygen -t rsa -P "", cat $HOME/.ssh_id.pub >> $HOME/.ssh/authorized_keys
ssh localhost

```
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ su students
Password:
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:/home/nitttr$ cd
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ ssh-keygen -t rsa -P ''
Generating public/private rsa key pair.
Enter file in which to save the key (/home/students/.ssh/id_rsa):
Created directory '/home/students/.ssh'.
Your identification has been saved in /home/students/.ssh/id_rsa
Your public key has been saved in /home/students/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:7Ji84gOWBSwwfV9FJy7vu4STq3b1KYoqWh7eSmf+/VA students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC
The key's randomart image is:
+---[RSA 3072]----+
|+o       o+ .    |
|..+ .   .. o     |
| . o . .. .      |
|     . .. o      |
|   o    S E      |
|  +  . + =.      |
| .+.o + *.o. .   |
| =.*o .+.=..o    |
|..++=*=o+o+o     |
+----[SHA256]-----+
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

```
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ ssh localhost
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

2 devices have a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.


Expanded Security Maintenance for Applications is not enabled.
```

Configuring the files.

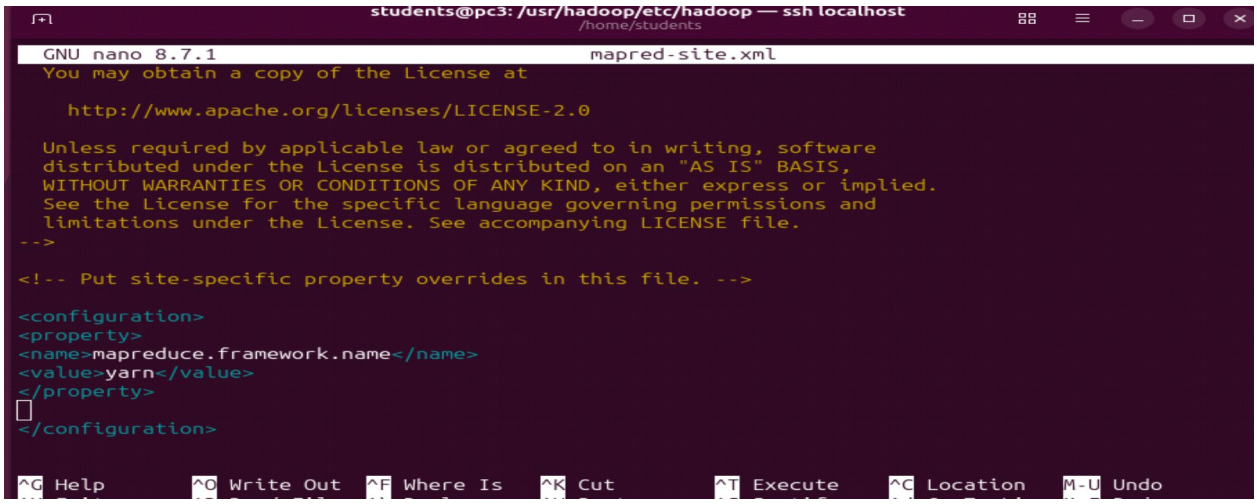
nano $HMOE/.bashrc

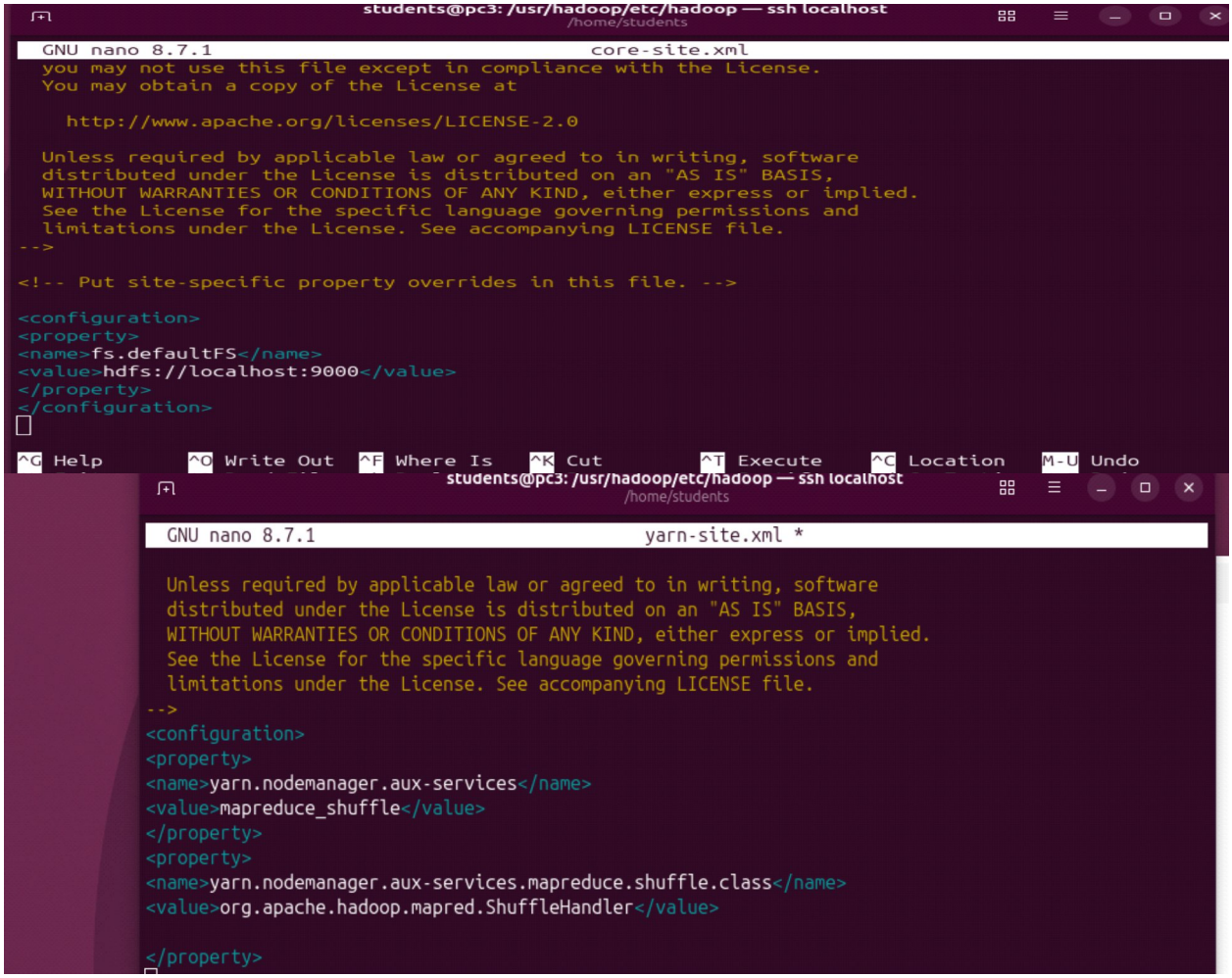
```
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi

export HADOOP_HOME=/usr/hadoop
export HADOOP_INSTALL=/usr/hadoop
export HADOOP_MAPRED_HOME=$HADOOP_INSTALL
export HADOOP_COMMON_HOME=$HADOOP_INSTALL
export HADOOP_YARN_HOME=$HADOOP_INSTALL
export HADOOP_HDFS_HOME=$HADOOP_INSTALL
export PATH=$PATH:$HADOOP_INSTALL/bin
export PATH=$PATH:$HADOOP_INSTALL/sbin
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute   ^C Location  M-U Undo
```

nano mapred-site.xml, nano core-site.xml, nano yarn-site.xml,





nano hdfs-site.xml

```
<configuration>
<property>
<name>dfs.replication</name>
<value>2</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>/data/hadoop/namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>/data/hadoop/datanode</value>
</property>
<property>
<name>dfs.permissions</name>
<value>false</value>
</property>
<property>
<name>dfs.datanode.use.datanode.hostname</name>
<value>false</value>
</property>
<property>
<name>dfs.namenode.datanode.registration.ip-hostname-check</name>
<value>false</value>
</property>
</configuration>
```

Source .bashrc , Format **hadoop namedone -format**
and start hsdoop services and verify they are running using **start-all.sh ,jps.**

```
2026-10-30 18:59:43,714 INFO namenode.NameNode: SHUTDOWN_MSG:
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at nitttr-HP-Elite-Tower-800-G9-Desktop-PC/127.0.1.1
************************************************************/
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as students in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [nitttr-HP-Elite-Tower-800-G9-Desktop-PC]
nitttr-HP-Elite-Tower-800-G9-Desktop-PC: Warning: Permanently added 'nitttr-hp-elite-tower-800-g9-de
sktop-pc' (ED25519) to the list of known hosts.
Starting resourcemanager
Starting nodemanagers
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ jps
103252 SecondaryNameNode
103015 DataNode
104118 Jps
103881 NodeManager
102828 NameNode
103535 ResourceManager
students@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

We have made single node. now we are move on to multinode and changes making in configuration files.
first we are set our hostname as pc3 **sudo hostnamectl set-hostname pc3, ifconfig**

```
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo hostnamectl set-hostname pc3
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ ifconfig
eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.16.2.192  netmask 255.255.255.0  broadcast 172.16.2.255
        ether 5c:60:ba:34:8d:f4  txqueuelen 1000  (Ethernet)
        RX packets 2384491  bytes 1867935156 (1.8 GB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 142712  bytes 18768590 (18.7 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 19  memory 0x82280000-822a0000

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 143984  bytes 13353932 (13.3 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 143984  bytes 13353932 (13.3 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

Go to superuser and add all nodes, the master inside our hosts file.The commond is **sudo nano /etc/hosts ,** then restart networkmanager:**sudo systemctl restart NetworkManager**

```
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC: ~ — ssh localhost
/home/students

nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo nano /etc/hosts
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ sudo systemctl restart NetworkManager
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ ping master
PING master (172.16.2.180) 56(84) bytes of data.
64 bytes from master (172.16.2.180): icmp_seq=1 ttl=64 time=1.71 ms
64 bytes from master (172.16.2.180): icmp_seq=2 ttl=64 time=0.912 ms
64 bytes from master (172.16.2.180): icmp_seq=3 ttl=64 time=1.12 ms
64 bytes from master (172.16.2.180): icmp_seq=4 ttl=64 time=0.796 ms
64 bytes from master (172.16.2.180): icmp_seq=5 ttl=64 time=0.886 ms
^C
--- master ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4040ms
rtt min/avg/max/mdev = 0.796/1.084/1.706/0.328 ms
nitttr@nitttr-HP-Elite-Tower-800-G9-Desktop-PC:~$ cat /etc/hosts
172.16.2.192 pc3
172.16.2.235 pc1
172.16.2.180 master

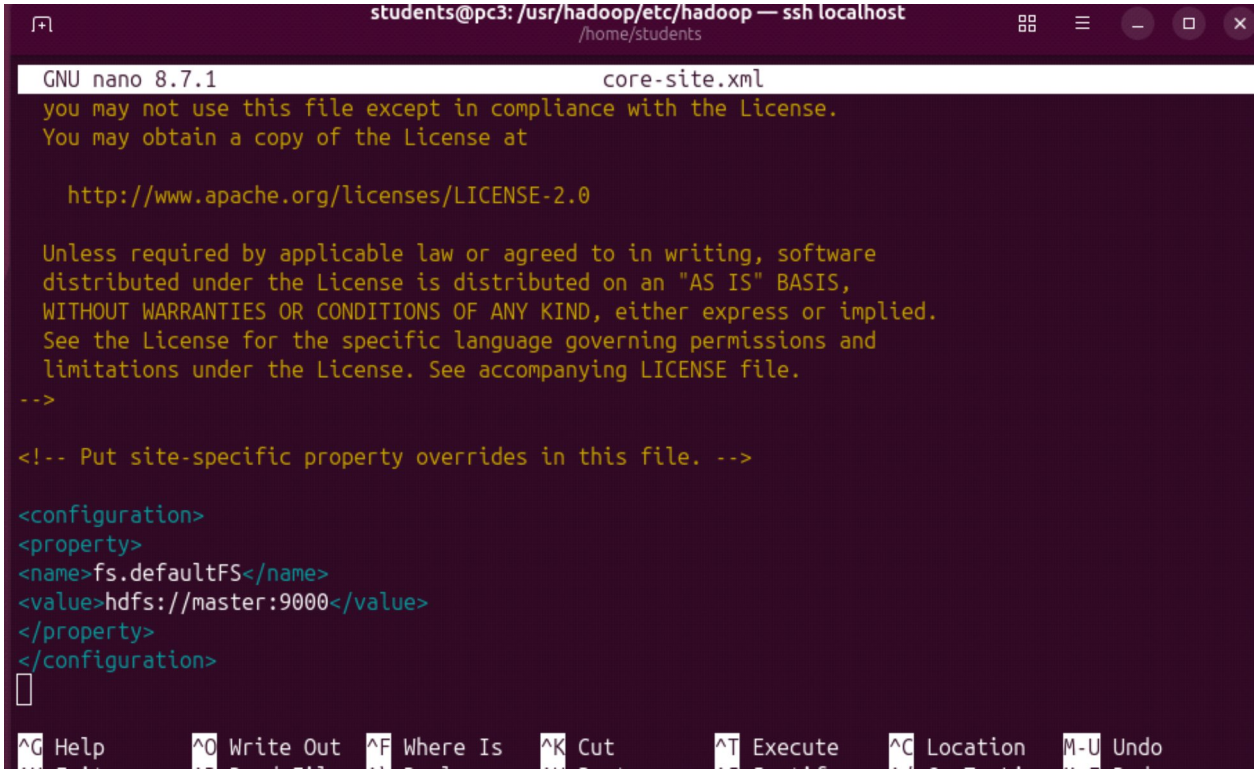
# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

then we make changes inside the **core-site.xml,mapred-site.xml,yarn-site.xml,hdfs-site.xml**

```
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
<name>mapreduce.framework.name</name>
<value>yarn</value>
</property>
<property>
<name>mapreduce.application.classpath</name>
<value>/usr/hadoop/share/hadoop/mapreduce/*:/usr/hadoop/share/hadoop/mapreduce/lib-examples/*</value>
</property>
</configuration>
```



```
<configuration>
<property>
<name>dfs.replication</name>
<value>2</value>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>/data/hadoop/namenode</value>
</property>
<property>
<name>dfs.datanode.data.dir</name>
<value>/data/hadoop/datanode</value>
</property>
<property>
<name>dfs.permissions</name>
<value>false</value>
</property>
<property>
<name>dfs.datanode.use.datanode.hostname</name>
<value>false</value>
</property>
<property>
<name>dfs.namenode.datanode.registration.ip-hostname-check</name>
<value>false</value>
</property>
</configuration>
```

students@pc3: /usr/hadoop/etc/hadoop — ssh localhost
/home/students

GNU nano 8.7.1 yarn-site.xml *

```
  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<configuration>
<property>
<name>yarn.nodemanager.aux-services</name>
<value>mapreduce_shuffle</value>
</property>
<property>
<name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
<value>org.apache.hadoop.mapred.ShuffleHandler</value>

</property>
<property>
<name>yarn.resourcemanager.hostname</name>
<value>master</value>
</property>
```

create the data folder and we changed namenode, datanoda,storages

```
students@pc3:~$ su nitttr
Password:
nitttr@pc3:/home/students$ cd
nitttr@pc3:~$ sudo mkdir -p /data/hadoop
nitttr@pc3:~$ sudo chown -R students:bdagirls /data
nitttr@pc3:~$
```

copy ssh keys of others nodes and master for running connection.

```
students@pc3:~$ ssh hduser@master
hduser@master's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-38-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

2 devices have a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.


Expanded Security Maintenance for Applications is not enabled.

22 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


2 devices have a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

Last login: Sat Oct 24 12:43:28 2026 from 172.16.2.235
hduser@master:~$
logout
Connection to master closed.
students@pc3:~$ ssh-copy-id hduser@master
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/students/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
hduser@master's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'hduser@master'"
and check to make sure that only the key(s) you wanted were added.

students@pc3:~$ ssh-copy-id hduser@pc1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/students/.ssh/id_rsa.pub"
The authenticity of host 'pc1 (172.16.2.235)' can't be established.
ED25519 key fingerprint is: SHA256:IO2JIR8oiEQjfB+k1ZEuvvFgOw1lB3jLKWDACdM9JYE
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
```

now start hadoop services, from nodes and the master and check they are running or not.

start-all.sh, jps

```
students@pc3:~$ jps
111440 DataNode
111560 NodeManager
111823 Jps
students@pc3:~$
```

B. Store Dataset in HDFS

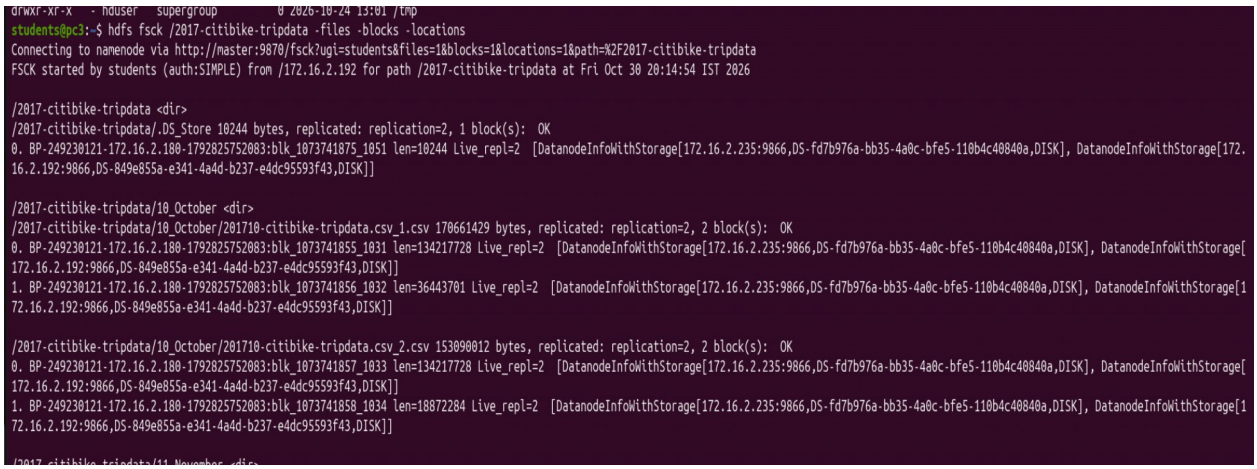
we have downloads 2017-citibike-tripdata, close to 500 megabytes. Added it to hdfs:**hdfs dfs -copyFromLocal /input/2017-citibike-tripdata /**

```
students@pc3:/home/nitttr$ cd
students@pc3:~$ hdfs dfs -copyFromLocal /input/2017-citibike-tripdata /
students@pc3:~$ hdfs dfs -ls /
Found 3 items
drwxr-xr-x   - students supergroup          0 2026-10-30 20:12 /2017-citibike-tripdata
drwxr-xr-x   - hduser   supergroup          0 2026-10-24 12:57 /cloud
drwxr-xr-x   - hduser   supergroup          0 2026-10-24 13:01 /tmp
```

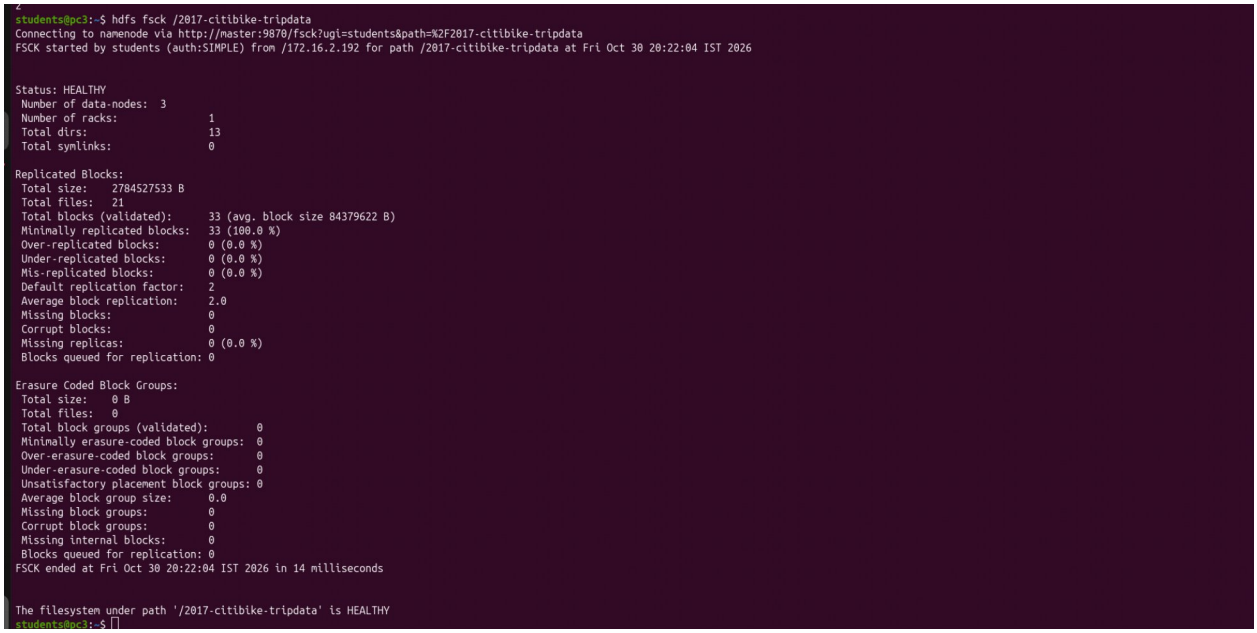
Task 3: Demonstrate Data Replication, Recovery & Management

chack the file metadata where number of blocks ars stored in, also the replication factor

hdfs fsck /2017-citibike-tripdata -files -blocks -locations



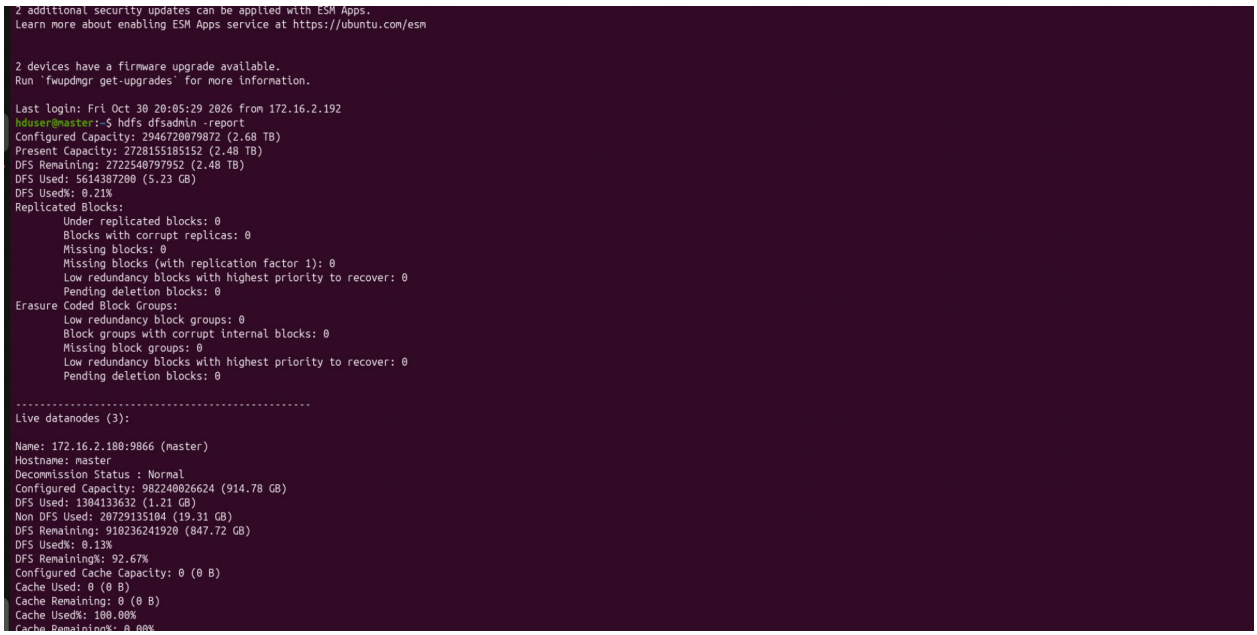
then verifying the file health:**hdfs fsck /2017-citibike-tridata**



The replication factor is 2, each block is stored in 2 copies on different DataNodes.This helps access data even if one DataNode fails. HDFS divides files into blocks (default block size is 128 MB) and stores them across different DataNodes.The NameNode checks DataNode availability, manages block locations, and maintains metadata. It does not store actual file blocks.The NameNode selects DataNodes to store multiple copies of each block and maintains the replication factor among the cluster.

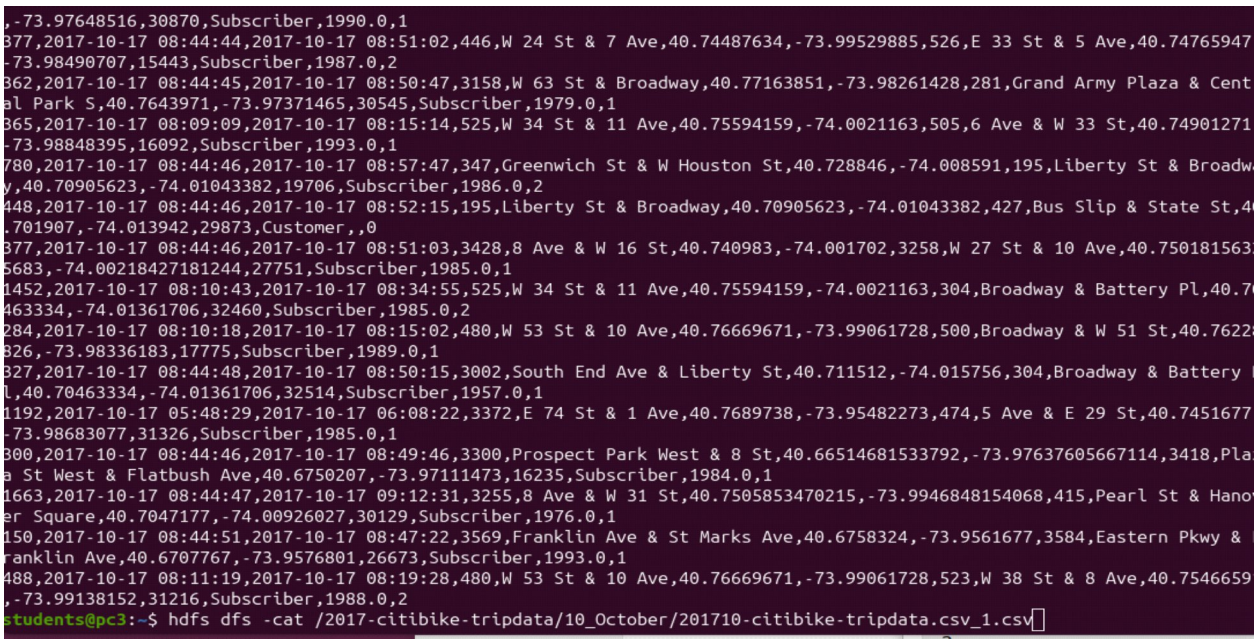
Task 4: Testing & Validation (Fault Tolerance Simulation)

let's check the number of datanodes running at the time:**hdfs dfsadmin -refort**

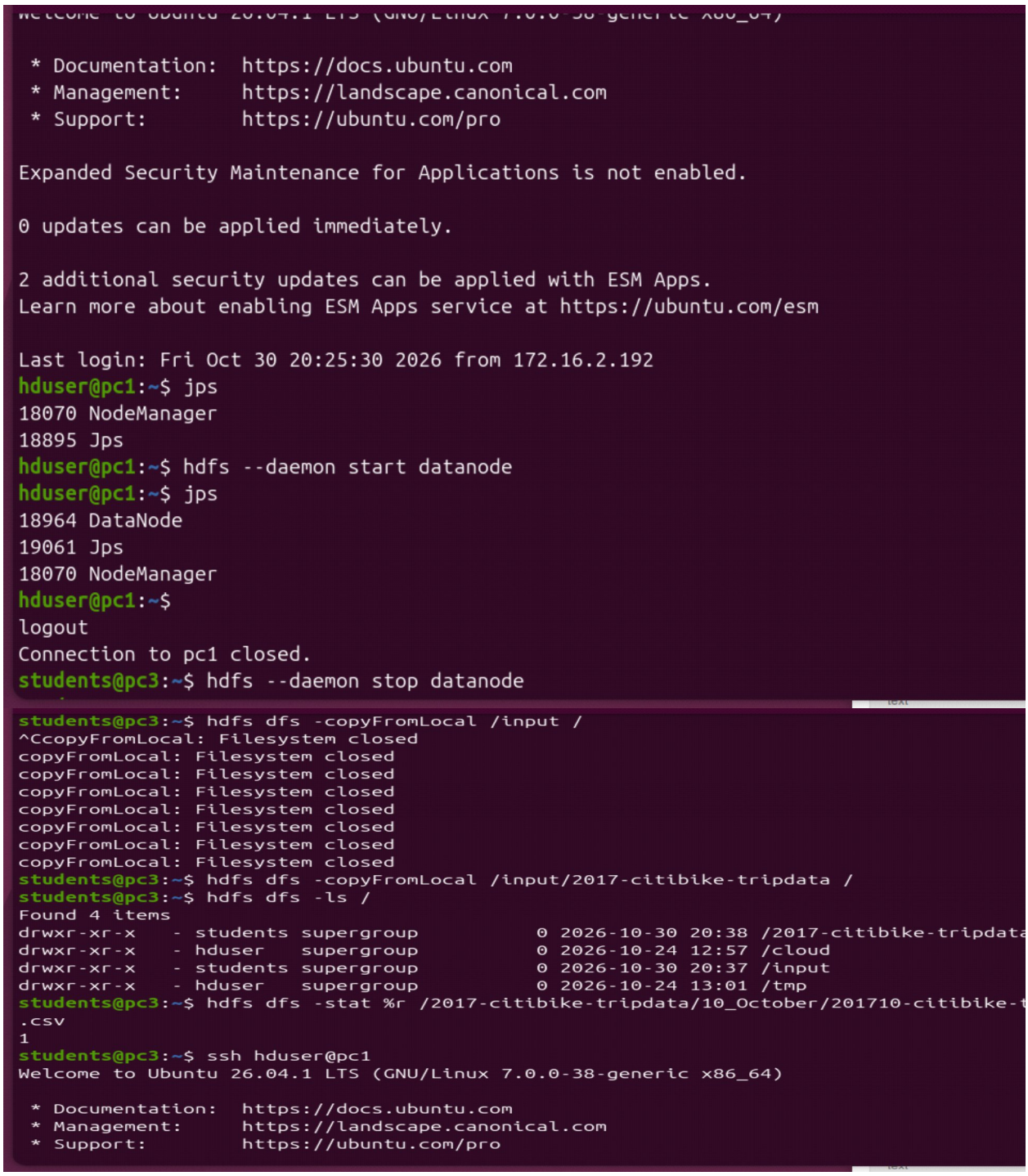




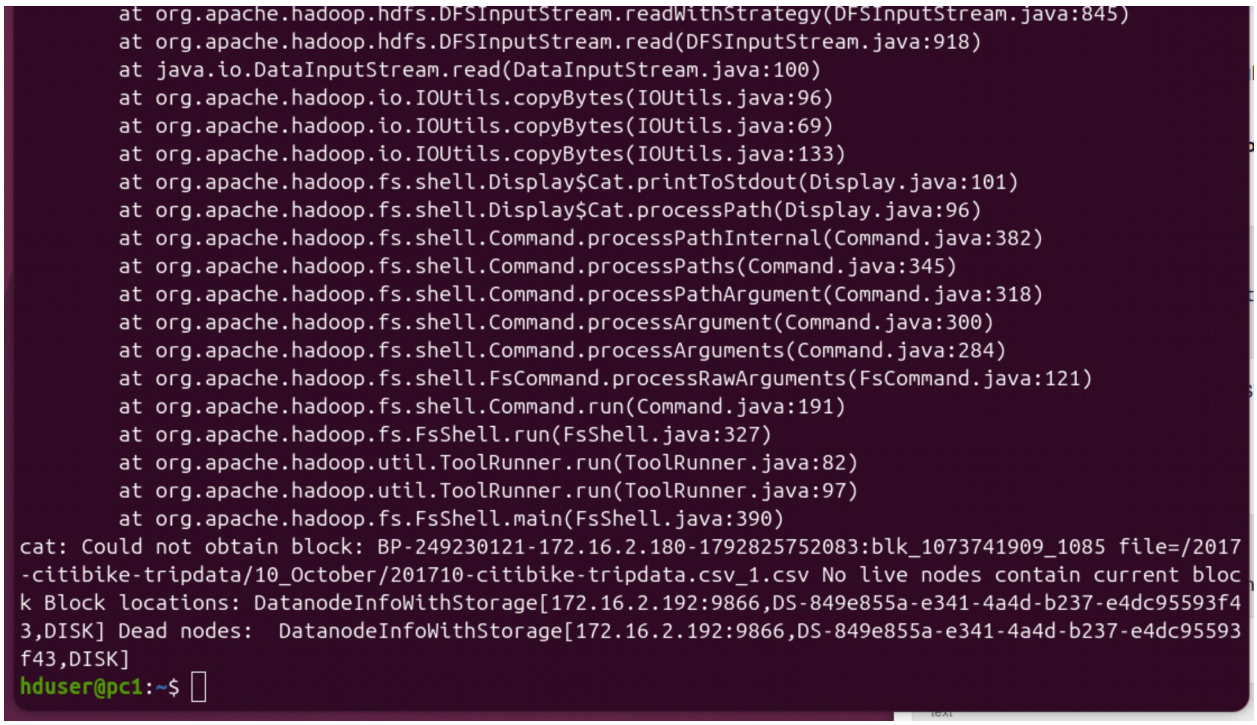
Now we have three datanodes running right here. sotp one datanode (in pc1).now access the file from pc3



The replication factor is 2 we could access the file,if when one datanode is off.make replication factor as a 1 and chack the fualt tolerance.



try to access the file now.



We could not access the block because the DataNode was down and the replication factor was one.

What I Have Learned:

Learned Hadoop installation and configuration.Understood how a multi-node HDFS cluster works.Learned about NameNode and DataNode communication.Understood HDFS fault tolerance through replication.Learned how replication keeps files accessible when a DataNode fails.Understood why HDFS is preferred for big data storage.